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Module Objectives

Discrete Logarithm Problem

Diffie-Hellman Key Exchange

- Video: Diffie-Hellman Key Exchange Overview
3 min
- Video: Diffie-Hellman Key-Exchange Protocol
3 min
- Reading: Diffie and Hellman, 1976
1h
- Video: Man-in-the-Middle Attack
3 min
- Video: El Gamal Encryption
1 min
- Reading: Lecture Slides for Diffie-Hellman Key Exchange
15 min
- Quiz: Diffie-Hellman Key Exchange and El Gamal Encryption
Submitted

🎉 Congratulations! You passed!

Grade received 83.33%

Latest Diffie-Hellman Key Exchange Grade 83.33%

To pass 80% or higher

Retake the assignment in 7h

Go to next

Diffie-Hellman Key Exchange and El Gamal Encryption

1. Alice and Bob exchanges a key using Diffie-Hellman Key Exchange protocol. Which of the followings do Alice know by the end of the protocol? Select all that applies. 4 / 4 points

Submit your assignment

Try again

Due Apr 28, 11:59 PM IST Attempts 3 every 8 hours

Retake the quiz in 7h 58m

☒ The prime modulus (p)

☒ Correct
See Diffie-Hellman Key Exchange Protocol lesson.
To Pass 80% or higher

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☒ The primitive root of the prime modulus (a)

☒ Correct
See Diffie-Hellman Key Exchange Protocol lesson.
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☒ The shared key by the protocol (K)

☒ Correct
See Diffie-Hellman Key Exchange Protocol lesson.

☐ Bob's private key (X_B)

☒ Bob's public key (Y_B)

☒ Correct
See Diffie-Hellman Key Exchange Protocol lesson.

2. Alice and Bob exchanges a key using Diffie-Hellman Key Exchange protocol. Which of the followings are to be secret against a passive attacker (who wishes to learn the shared key)? Select all that applies. 0 / 4 points

☒ Bob's private key (Y_B)

☒ Correct

☐ The prime modulus (p)

☐ The primitive root of the prime modulus (a)

☐ Bob's public key (X_B)

☐ The shared key by the protocol (K)

You didn't select all the correct answers

3. Alice and Bob uses Diffie-Hellman Key Exchange to share a key with a common prime p=11 and a primitive root a=2. If Alice has a public key Y_A=9, what is her private key X_A? 4 / 4 points

6

☒ Correct
2^6 mod 11 = 9

4. (This question builds on the previous question.) Alice and Bob uses Diffie-Hellman Key Exchange to share a key with a common prime p=11 and a primitive root a=2. If Bob has a public key Y_B=3, what is the secret key K shared with Alice? 4 / 4 points

3

☒ Correct
3^6 mod 11 = 3 computed by Alice

5. Which of the followings are true for Diffie-Hellman Key Exchange and Man-in-the-middle (MITM) attack? Select all that applies. 4 / 4 points

☐ MITM attack is a passive attack.

☐ Diffie-Hellman Key Exchange protocol is widely used for digital signature.

☐ Diffie-Hellman Key Exchange protocol establishes a shared secret key between the two parties involved in the protocol and therefore is considered symmetric cryptography.

☒ Diffie-Hellman Key Exchange protocol is vulnerable to MITM attack because of the lack of authentication.

☒ Correct

6. Which of the followings are true about El Gamal Encryption? 4 / 4 points

☒ El Gamal Encryption uses a prime modulus and a primitive root of the modulus.

☒ Correct

☐ El Gamal Encryption relies on prime factorization problem.

☒ El Gamal Encryption includes the message and the key that is used to protect the message, and the key itself is protected against eavesdropping.

☒ Correct

☐ El Gamal Encryption is used for key exchange between the participating parties.